

Zytel® 70G33L NC010

NYLON RESIN

Product Description

Zytel® 70G33L NC010 is a 33% glass fiber reinforced polyamide 66 resin for injection molding.

General

Material Status	• Commercial: Active		
Availability	• Africa & Middle East • Asia Pacific • Central America	• Europe • Latin America • North America	• South America
Filler / Reinforcement	• Glass Fiber Reinforcement, 33% Filler by Weight		
Features	• General Purpose • Good Creep Resistance	• Lubricated • Ultrasonic Weldable	
Uses	• Appliance Components • Automotive Applications	• Electrical/Electronic Applications • Industrial Applications	
RoHS Compliance	• Contact Manufacturer		
Appearance	• Natural Color		
Forms	• Pellets		
Processing Method	• Injection Molding		
Multi-Point Data	• Isothermal Stress vs. Strain (ISO 11403-1)		
Part Marking Code (ISO 11469)	• >PA66-GF33<		
Resin ID (ISO 1043)	• PA66-GF33		
Product Category	• Glass Reinforced Resins		

Physical	Dry	Conditioned	Unit	Test Method
Specific Gravity				
--	1.38	--	g/cm ³	ASTM D792
--	1.39	--	g/cm ³	ISO 1183
Molding Shrinkage				
Flow: 1.60 mm	0.20	--	%	Internal Method
Flow: 3.20 mm	0.30	--	%	Internal Method
Flow: 6.40 mm	0.50	--	%	Internal Method
Across Flow: 1.60 mm	1.0	--	%	Internal Method
Across Flow: 3.20 mm	1.0	--	%	Internal Method
Across Flow: 6.40 mm	1.1	--	%	Internal Method
Across Flow: 2.00 mm	1.1	--	%	ISO 294-4
Flow: 2.00 mm	0.30	--	%	ISO 294-4
Water Absorption				
23°C, 24 hr	0.70	--	%	ASTM D570
23°C, 24 hr	1.2	--	%	ISO 62
Saturation, 23°C	5.4	--	%	ASTM D570
Saturation, 23°C	5.7	--	%	ISO 62
Equilibrium, 23°C, 50% RH	1.8	--	%	ISO 62
Mechanical	Dry	Conditioned	Unit	Test Method
Tensile Modulus (23°C)	10500	8000	MPa	ISO 527-2
Tensile Stress				
Break, 23°C	200	140	MPa	ISO 527-2
-40°C	214	207	MPa	ASTM D638
23°C	186	124	MPa	ASTM D638
77°C	110	86.0	MPa	ASTM D638

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Mechanical	Dry	Conditioned	Unit	Test Method
Tensile Elongation				
Break, 23°C	3.0	4.0	%	ASTM D638
Break, 23°C	3.5	5.0	%	ISO 527-2
Flexural Modulus				
23°C	8970	6210	MPa	ASTM D790
23°C	9300	6210	MPa	ISO 178
Flexural Strength				
23°C	262	--	MPa	ASTM D790
23°C	290	200	MPa	ISO 178
Shear Strength (23°C)	86.0	--	MPa	ASTM D732
Impact	Dry	Conditioned	Unit	Test Method
Charpy Notched Impact Strength				ISO 179/1eA
-40°C	10	10	kJ/m ²	
-30°C	10	10	kJ/m ²	
23°C	13	17	kJ/m ²	
Charpy Unnotched Impact Strength				ISO 179/1eU
-30°C	70	75	kJ/m ²	
23°C	85	100	kJ/m ²	
Notched Izod Impact				
23°C	120	130	J/m	ASTM D256
-40°C	10	10	kJ/m ²	ISO 180/1A
-30°C	10	10	kJ/m ²	ISO 180/1A
23°C	12	15	kJ/m ²	ISO 180/1A
Unnotched Izod Impact				
23°C	1300	1500	J/m	ASTM D4812
-30°C	70	70	kJ/m ²	ISO 180/1U
23°C	80	90	kJ/m ²	ISO 180/1U
Hardness	Dry	Conditioned	Unit	Test Method
Rockwell Hardness (M-Scale)	101	--		ASTM D785
Thermal	Dry	Conditioned	Unit	Test Method
Deflection Temperature Under Load				
0.45 MPa, Unannealed	261	--	°C	ASTM D648
1.8 MPa, Unannealed	249	--	°C	ISO 75-2/B
1.8 MPa, Unannealed	252	--	°C	ASTM D648
Glass Transition Temperature	80.0	--	°C	ISO 75-2/A
Melting Temperature				
-- ²	262	--	°C	ISO 11357-3
--	263	--	°C	ASTM D3418
CLTE				ISO 11359-2
Flow: -40 to 23°C	0.000024	--	cm/cm/°C	
Flow: 23 to 55°C	0.000018	--	cm/cm/°C	
Flow: 55 to 160°C	0.000013	--	cm/cm/°C	
Transverse: -40 to 23°C	0.000065	--	cm/cm/°C	
Transverse: 23 to 55°C	0.000083	--	cm/cm/°C	
Transverse: 55 to 160°C	0.00014	--	cm/cm/°C	
RTI Elec (0.710 mm)	130	--	°C	UL 746
RTI Imp (0.710 mm)	120	--	°C	UL 746
RTI Str (0.710 mm)	130	--	°C	UL 746
Electrical	Dry	Conditioned	Unit	Test Method
Volume Resistivity				
23°C	1.0E+15	--	ohm·cm	ASTM D257
--	1.0E+15	--	ohm·cm	IEC 60093

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Electrical	Dry	Conditioned	Unit	Test Method
Dielectric Strength ³ (23°C, 3.20 mm)	21	--	kV/mm	ASTM D149
Dielectric Constant				
23°C, 1 kHz	4.50	--		ASTM D150
23°C, 1 MHz	3.70	--		ASTM D150
23°C, 100 Hz	4.20	--		IEC 60250
23°C, 1 MHz	4.00	--		IEC 60250
Dissipation Factor				
23°C, 1 kHz	0.020	--		ASTM D150
23°C, 1 MHz	0.020	--		ASTM D150
23°C, 100 Hz	0.010	--		IEC 60250
23°C, 1 MHz	0.015	--		IEC 60250
Comparative Tracking Index (CTI)	600	--	V	UL 746
Comparative Tracking Index	600	--	V	IEC 60112
Flammability	Dry	Conditioned	Unit	Test Method
Flame Rating (0.710 mm)	HB	--		UL 94
Oxygen Index	24	--	%	ISO 4589-2
Injection		Dry Unit		
Drying Temperature		80.0 °C		
Drying Time		2.0 to 4.0 hr		
Suggested Max Moisture		< 0.20 %		
Processing (Melt) Temp		285 to 305 °C		
Melt Temperature, Optimum		295 °C		
Mold Temperature		70.0 to 120 °C		
Mold Temperature, Optimum		100 °C		
Drying Recommended		Yes, if moisture content of resin exceeds recommended level		

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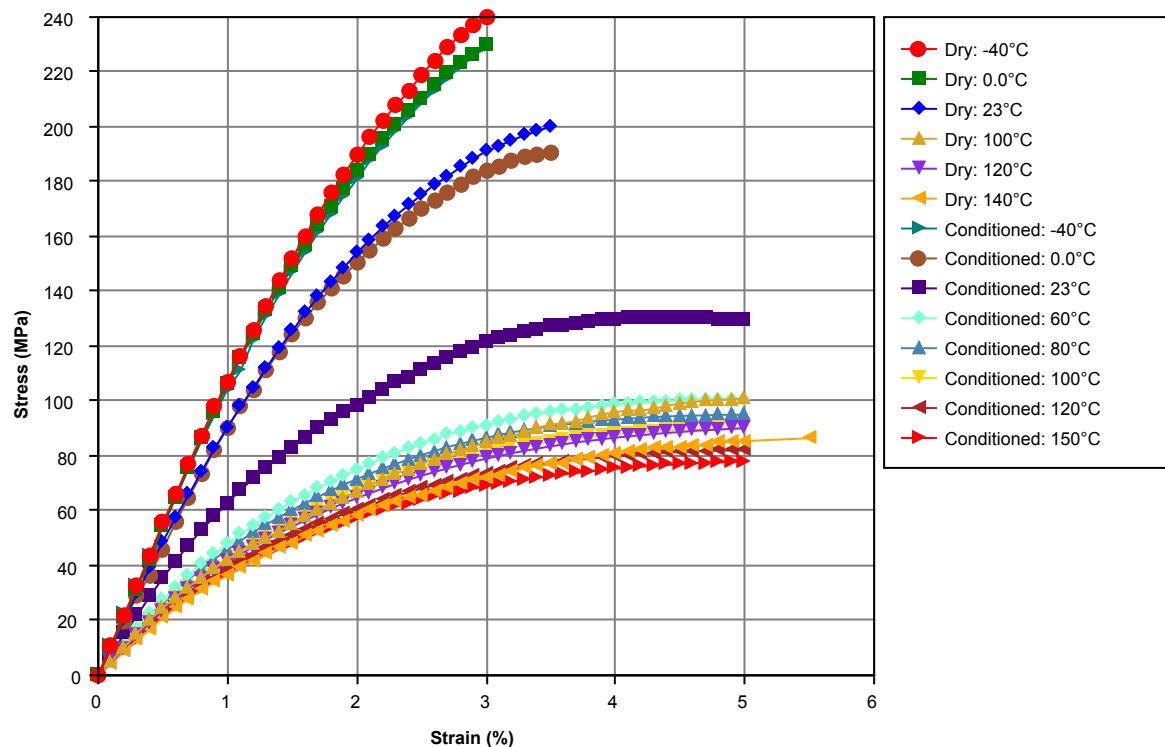
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Isothermal Stress vs. Strain (ISO 11403-1)



Notes

¹ Typical properties: these are not to be construed as specifications.

² 10°C/min

³ Method A (Short-Time)

Contact DuPont for Material Safety Data Sheet, general guides and/or additional information about ventilation, handling, purging, drying, etc.

ISO Mechanical properties measured at 4.0mm, ISO Electrical properties measured at 2.0mm, and all ASTM properties measured at 3.2mm.

Test temperatures are 23°C unless otherwise stated.

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